Calibration Bets on the Past: Post-Training Quantization for Financial Time-Series Forecasting

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Abstract

Financial forecasting models are typically developed in full precision, yet production deployment often requires low-precision inference to reduce memory and computational cost. Post-training quantization (PTQ) enables such deployment without retraining. However, reliable activation quantization requires calibration: activation ranges are estimated from historical data before deployment and then remain fixed during future inference. We systematically study activation calibration for PTQ in cross-sectional volatility forecasting on the S&P 500. We evaluate seven neural architectures over eight walk-forward test years (2018–2025), training 560 models. We find that activation calibration has little effect at 8 bits but becomes the primary determinant of predictive performance at 4 bits. Under default absolute-maximum (abs-max) calibration, static 4-bit quantization of both weights and activations removes 11–62% of the full-precision mean information coefficient in affected architectures. Replacing abs-max with the best-tested percentile recovers up to 53–94% of this degradation in the four most affected architectures, although the recurrent SegRNN and the multi-scale TimeMixer retain substantial residual losses. The preferred activation range also varies across market periods. Narrow ranges improve resolution under typical market conditions but lose part of their advantage when test-period market dispersion exceeds the calibration history. These findings show that activation calibration is a first-class deployment decision for reliable 4-bit PTQ in financial forecasting. When substantial degradation remains, 8-bit activations or weight-only 4-bit quantization are more robust alternatives.

Code Availability: https://github.com/JunyiYe/calibration-bets-ptq

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Machine learning*; • **Mathematics of computing** → **Time series analysis**; • **Applied computing** → *Economics*.

Keywords

post-training quantization, activation calibration, distribution shift, financial time-series forecasting, volatility forecasting



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ICAIF '26, Milan, Italy
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ACM ISBN 979-8-4007-3087-0/2026/11
https://doi.org/10.1145/3858063.3858796

ACM Reference Format:
Junyi Ye and Ivy Gateri Wanjiku. 2026. Calibration Bets on the Past: Post-Training Quantization for Financial Time-Series Forecasting. In *7th ACM International Conference on AI in Finance (ICAIF '26), November 14–17, 2026, Milan, Italy.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3858063.3858796

1 Introduction

Deploying financial forecasting models in production often requires low-precision inference to satisfy practical memory, latency, and hardware constraints [3, 8, 16, 18]. These constraints become more consequential when models repeatedly generate forecasts across large asset universes, because inference costs accumulate across assets and update cycles. Post-training quantization (PTQ) provides an attractive deployment strategy by converting a trained model into a low-precision model without retraining. However, the different components of PTQ do not contribute equally to deployment accuracy. In practice, weight quantization is often robust, whereas activation quantization depends critically on how activation ranges are calibrated before deployment.

Unlike model weights, activation ranges cannot be computed directly from trained parameters. Instead, they are estimated from a historical calibration dataset before deployment and then remain fixed throughout future inference. This makes activation calibration a deployment decision rather than merely a numerical implementation detail. A wide range preserves rare activation extremes but reduces numerical resolution for typical activations, whereas a narrow range improves resolution while increasing clipping. In financial forecasting, this trade-off is particularly challenging because calibration always relies on historical market data, whereas deployment occurs under future market conditions that may differ substantially from the calibration period.

Existing PTQ research has developed effective calibration methods, outlier-handling techniques, and low-bit quantization schemes for vision and language models. However, relatively little is known about the predictive consequences of activation calibration in financial forecasting. In particular, it remains unclear how sensitive different forecasting architectures are to activation calibration, how much of the resulting degradation can be recovered through improved range selection, and when calibration itself becomes a limiting factor for low-precision deployment.

To answer these questions, we systematically evaluate activation calibration under a walk-forward PTQ protocol for cross-sectional volatility forecasting. The study spans seven representative neural architectures, eight walk-forward test years from 2018 to 2025, and 560 independently trained models on an S&P 500 equity panel. By

comparing each quantized model against its own full-precision checkpoint while varying only the activation calibration strategy, we isolate the predictive impact of activation-range selection.

Our study leads to four main observations. First, 8-bit quantization and 4-bit weight-only quantization introduce little predictive degradation for most architectures. Second, quantizing activations to 4 bits causes the largest losses, with affected models losing 11–62% of their full-precision mean information coefficient under the default range-selection strategy. Third, much of this loss is recoverable through improved range selection, although recurrent and multi-scale architectures retain substantial residual sensitivity. Finally, the preferred activation range itself changes over time, reinforcing the view that activation calibration should be treated as a deployment decision rather than a fixed preprocessing step.

The contributions of this paper are as follows.

- To the best of our knowledge, this is the first systematic study of activation calibration for post-training quantization in financial forecasting under a walk-forward evaluation protocol.
- We reveal that 4-bit activation degradation consists of both range-recoverable loss and architecture-dependent residual loss: improved calibration recovers much of the degradation in several architectures, while recurrent and multi-scale models remain substantially sensitive.
- We show that activation-range preferences evolve across market periods and translate these observations into practical deployment guidelines for selecting among percentile W4A4, layer-wise mixed precision, W8A8, and weight-only W4 quantization.

2 Related Work

Our work connects two research directions: activation calibration for PTQ and low-precision inference for time-series forecasting.

Activation calibration for PTQ. Weight quantization is generally robust because weight ranges can be computed directly from trained parameters. Activation quantization is more challenging because activation ranges must be estimated from a calibration dataset before deployment. Existing PTQ research has developed calibration strategies, clipping methods, and quantization-aware activation functions for low-bit inference [2, 5]. More recent work shows that activation outliers dominate many low-bit failures in large language models, motivating mixed-precision decomposition [7], channel-wise scaling [22], and activation rotations [1]. Calibration data themselves also influence PTQ accuracy even when the trained model and compression algorithm remain fixed [21]. However, these studies focus primarily on vision and language models.

Low-precision time-series forecasting. Low-precision inference has also been studied for sequential and time-series models through retraining [17] and quantization-aware training [3, 9]. More recently, Pavlova et al. [16] show that PTQ sensitivity varies across time-series models and use this variation to motivate mixed-precision allocation. However, these studies do not systematically investigate activation calibration or the selection of static activation ranges from calibration data. Most existing PTQ benchmarks assume calibration and evaluation data are drawn from the same underlying distribution [23], whereas financial forecasting requires calibration on historical data followed by deployment on future observations.

To the best of our knowledge, no prior work has systematically studied activation calibration for PTQ in financial forecasting; we characterize this trade-off under financial temporal shifts.

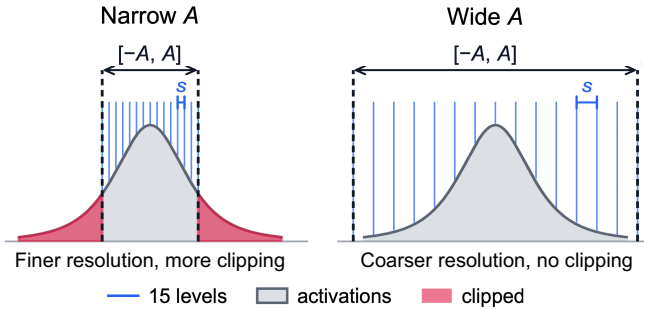


Figure 1: The resolution–range trade-off at 4 bits. Both panels show the same activation distribution quantized with the 15 representable values; only the range A differs. A narrow A reduces the step size s but clips the tails, whereas a wide A eliminates clipping at the cost of coarser spacing throughout.

3 Methodology

3.1 Low-bit activation quantization

Post-training quantization (PTQ) converts a trained full-precision neural network into a low-precision model without retraining, replacing floating-point weights and activations with a small set of discrete values to reduce memory usage and inference cost. The reduction in precision is particularly significant at low bit widths: under the symmetric 4-bit quantizer used in this paper, only 15 representable levels cover the entire numerical range of an activation tensor, so selecting this range is a central design decision.

The range–resolution trade-off can be understood from a standard symmetric quantizer [10, 14]. Let x denote a full-precision scalar and $\hat{x}$ its quantized–dequantized approximation. For a quantization range $[-A, A]$, we compute

$$\hat{x} = s \cdot \text{round}\left(\frac{\text{clip}(x, -A, A)}{s}\right), \qquad s = \frac{A}{2^{b-1} - 1}, \tag{1}$$

where s is the quantization step size for bit width b. For the 4-bit case, $s = A/7$.

Equation (1) illustrates the fundamental trade-off. A larger range reduces clipping by covering more extreme values but increases the step size, producing larger rounding errors. A smaller range improves numerical resolution while clipping more activations. The optimal range therefore balances clipping and rounding errors [2]. Even the optimal range generally cannot eliminate quantization error entirely, leaving a residual error that depends on both the model and the activation distribution. Figure 1 illustrates this trade-off by varying only the range parameter A while holding the activation distribution and quantization precision fixed.

3.2 Activation calibration

The quantization range must be determined separately for weights and activations. Weight tensors are fixed after training, so their ranges can be computed directly from the trained parameters. Activation values, however, depend on the input data and therefore require a calibration dataset before deployment.

We consider static PTQ, a widely used deployment setting in which activation ranges are calibrated once before deployment and then kept fixed throughout subsequent inference. Let C denote the calibration set. The activation range is estimated as

$$A = \text{stat}\left(\{|x| : x \in C\}\right), \tag{2}$$

where $\text{stat}(\cdot)$ denotes the calibration statistic. In our walk-forward evaluation, C is sampled from validation year $Y-1$, and the resulting range is applied unchanged throughout test year Y.

We evaluate two widely used calibration strategies. **Absolute-maximum (abs-max) calibration** selects

$$A = \max\left(\{|x| : x \in C\}\right),$$

thereby covering the full magnitude range observed during calibration. However, a single extreme activation can substantially enlarge the quantization range and reduce numerical resolution for typical activations. **Percentile calibration** instead selects

$$A = P_p(\{|x| : x \in C\}),$$

where P_p denotes the p-th percentile of activation magnitudes.

4 Experimental Setup

4.1 Dataset and sample construction

We construct a daily panel of the 501 current S&P 500 constituents from June 2008 through December 2025, comprising approximately 2.1 million asset-days. Daily simple returns are computed from Yahoo Finance[1] adjusted close prices. Five tickers renamed after the sample period broke their historical Yahoo Finance identifiers at retrieval time. Their histories are backfilled from Tiingo[2], with agreement within 10^{-3} over overlapping periods.

Each stock contributes eleven input features comprising lagged return statistics, technical indicators, and rolling distribution summaries: daily return; 5-, 10-, and 20-day mean returns; 20-day volatility; 14-day Relative Strength Index (RSI); Moving Average Convergence Divergence (MACD), computed as the difference of 12- and 26-day exponential moving averages (EMAs), and its 9-day signal line; and the 20-day maximum, minimum, and skewness of returns. Each feature is standardized using statistics estimated from the training split only.

Each input sample consists of a 64-day history of the eleven input features for a single stock. Models process stocks independently and produce one volatility forecast for each stock and forecast date.

4.2 Forecasting task and metric

We study cross-sectional prediction of five-day-forward volatility. For stock i at forecast origin t, define

$$v_{i,t} = \text{sd}\left(r_{i,t+1}, \ldots, r_{i,t+5}\right).$$

The supervised target is the within-date cross-sectional z-score of $\log v_{i,t}$, denoted by $y_{i,t}$. Cross-sectional volatility forecasting is relevant to risk ranking, position sizing, and volatility-aware portfolio construction [13]. Compared with daily return prediction, volatility provides a stronger predictive signal in this panel, making it better suited for measuring small quantization effects.

On each forecast date, the model scores all available stocks. Forecast skill is measured by the daily information coefficient (IC)

$$\text{IC}_t = \text{Spearman}_i\left(\hat{y}_{i,t}, y_{i,t}\right),$$

the cross-sectional Spearman correlation between predicted and realized scores. IC ranges from -1 to 1, with higher values indicating more accurate cross-sectional ranking and values near zero indicating no predictive skill. Reported IC values are averaged over all test dates. Section 4.7 builds on this metric to define quantization damage as the paired difference between an architecture's full-precision and quantized IC.

4.3 Walk-forward protocol

Evaluation follows a fixed-length walk-forward protocol. For test year Y, training uses the seven years $Y-8$ through $Y-2$, validation uses year $Y-1$, and testing uses year Y. Holding the training-window length fixed across folds ensures that comparisons across folds are not confounded by differences in training-window length. The test years span 2018–2025, producing eight folds. Each of the seven architectures is trained with ten random seeds per fold, yielding $8 \times 10 \times 7 = 560$ independent training runs.

To prevent information leakage across split boundaries, a sample is assigned to a split according to the final date of its five-day outcome window, $t+5$, where t is the forecast date, rather than by t itself. This guarantees that every return used to construct a validation or test target falls strictly within that sample's assigned split. The 64-day input window, by contrast, is allowed to extend backward across a split boundary, since those earlier observations are legitimately available at forecast time. Because input windows only reach backward while target windows are never split across a boundary, validation-year early stopping never has access to any test-year return, whether as a feature or as a label.

Hyperparameters are selected once using a development pseudo-fold with the same temporal structure (training: 2009–2015; validation: 2016), and the resulting configuration is frozen for all eight evaluation folds.

4.4 Models and training

We evaluate seven architectures spanning linear, mixer, attention, and recurrent time-series models: DLinear [24], TSMixer [4], TimeMixer [20], a vanilla Transformer [19], PatchTST [15], iTransformer [12], and SegRNN [11]. TSMixer uses MLP-based mixing, and TimeMixer uses MLP-based multi-scale mixing. Among the attention-based models, the vanilla Transformer applies attention directly over raw time steps without patching or channel tokenization, PatchTST applies attention over temporal patches, and iTransformer treats input channels as tokens. SegRNN applies recurrent processing over temporal segments. For each tunable architecture, we perform an equal-budget grid search on the development split over

$$d_{\text{model}} \in \{16, 32, 64, 128, 256\}, \qquad \text{dropout} \in \{0.1, 0.3, 0.5\}.$$

[1] https://finance.yahoo.com
[2] https://www.tiingo.com

Table 1: Precision settings evaluated in this work. Dynamic INT8 is included only as a reference.

Treatment	Weights	Activations	Activation calibration
FP32	FP32	FP32	–
W8A8	INT8	INT8	Static
W4	INT4	FP32	–
W4A4	INT4	INT4	Static
Dynamic INT8	INT8	Runtime INT8	Dynamic

Model depth is fixed at two, with the feed-forward dimension set to $d_{\text{ff}} = 4d_{\text{model}}$ when applicable. Each configuration is trained with three random seeds. The configuration achieving the lowest average validation mean squared error (MSE) is selected. The selected hyperparameters are then fixed for all walk-forward folds. DLinear has no tunable width, depth, or dropout under its published configuration and is therefore excluded from the grid search.

All models are trained with AdamW using a batch size of 1024 to minimize the MSE on the standardized target. Training uses early stopping on validation MSE with patience 20 and a maximum of 300 epochs, and the checkpoint with the lowest validation MSE is restored.

4.5 Classical baselines

We include two classical volatility-forecasting baselines as reference models. The first is a persistence forecast that carries each stock's trailing five-day realized volatility forward unchanged. The second is a pooled heterogeneous autoregressive (HAR) model [6] using daily, weekly, and monthly realized volatility as predictors. HAR is fitted using the training period of each fold and then applied unchanged to the corresponding validation and test periods.

4.6 PTQ protocol

Table 1 summarizes the evaluated precision settings. We refer to each precision setting as a *treatment*. FP32 serves as the full-precision reference. W4 isolates weight quantization by using INT4 weights with FP32 activations. W8A8 and W4A4 quantize both weights and activations under the static calibration protocol described in Section 3. Dynamic INT8 is included as a complementary reference. Unlike the static settings, it recomputes activation ranges during inference rather than relying on a fixed calibration range.

The primary experiments focus on W8A8, W4, and W4A4. These settings enable controlled comparisons of weight-only quantization, 8-bit activation quantization, and 4-bit activation quantization under a common static PTQ framework, isolating the quantization configuration and activation-calibration strategy from the underlying execution backend.

Activation calibration. For W8A8 and W4A4, activation ranges are estimated from 1,024 sequence windows sampled uniformly from the validation split preceding each test year. Unless otherwise specified, the resulting ranges remain fixed throughout the corresponding test year.

Quantization coverage. We quantize every learned matrix multiplication together with the activation presented to it, including linear projections, convolutional token embeddings, and recurrent matrix products. For SegRNN, the GRU is unrolled and both the step input and hidden-state operands are quantized at every time step. Biases, normalization layers, and elementwise nonlinearities remain in FP32.

For attention-based models, learned projection matrices and their input activations are quantized, whereas activation-only attention products remain in FP32. By quantizing every learned matrix multiplication regardless of architecture, we ensure that quantization coverage is comparable across models. Restricting quantization to `nn.Linear` modules would, for example, leave most of SegRNN's recurrent core unquantized.

Execution. W8A8, W4A4, and W4 are implemented using simulated quantization. Quantized operands pass through the quantize–dequantize transformation in Eq. (1), while matrix multiplication and accumulation are performed in FP32.

4.7 Quantization damage estimates

Quantization is applied to a fixed checkpoint, so FP32 and quantized forecasts are paired on the same stocks and dates. For architecture m, fold Y, seed s, treatment q, and test date t, define

$$\Delta_{m,Y,s,q,t} = \mathrm{IC}^{\mathrm{FP32}}_{m,Y,s,t} - \mathrm{IC}^{q}_{m,Y,s,t}.$$

We refer to $\Delta_{m,Y,s,q,t}$ as the *quantization damage*: positive values indicate that quantization reduces predictive skill relative to the full-precision checkpoint.

We summarize $\Delta_{m,Y,s,q,t}$ descriptively: it is first averaged over test dates within each fold-by-seed cell, and we then report the mean and standard deviation of the resulting fold-by-seed cell means (Table 2, discussed in Section 5). These summaries are purely descriptive and carry no significance markers. Throughout the paper, we interpret quantization damage through effect size, comparing it against the HAR and persistence baselines, rather than through formal hypothesis testing.

5 Main Results

Table 2 summarizes the predictive effects of each quantization setting across 560 walk-forward training runs. The first three quantized settings (Dynamic INT8, W8A8, and W4) serve as reference configurations, whereas the remaining columns compare static W4A4 under different activation-calibration strategies. HAR and persistence provide classical forecasting baselines that help interpret the practical magnitude of the observed IC losses.

Result 1: Dynamic INT8, W8A8, and W4 preserve nearly all predictive skill. The three reference quantization settings introduce only minor predictive damage. Under static W8A8, six of seven architectures lose no more than 0.0007 mean daily IC. For five of these six, this corresponds to less than 0.2% of their FP32 signal. The exception is DLinear, whose near-zero FP32 IC (0.026) turns the same small absolute change into a disproportionately large percentage (2.7%). Dynamic INT8, which recomputes activation ranges during inference, is similarly stable: for six architectures, its damage differs from static W8A8 by less than 0.001 IC. TimeMixer is the only model with a clearly visible 8-bit effect, and its loss is smaller under dynamic than static quantization.

Weight-only W4 is likewise robust in absolute terms. Transformer and TSMixer lose at most 0.5% of their FP32 IC. DLinear and

Table 2: Quantization damage (Δ; lower is better) across architectures. Entries report mean ± s.d. across eight test years and ten seeds per year; parentheses give damage as a share of FP32 IC where meaningful. The abs-max, p99.9, and p99 columns are static W4A4 settings differing only in the activation range statistic, with the lowest-damage setting per architecture shown in bold. R and L (Section 6.2) use the best setting in the full Figure 2 sweep, which can be slightly lower than the bold entries.

			Reference settings			Static W4A4 activation range		
Model	Family	FP32 IC ↑	ΔDyn8 ↓	ΔW8A8 ↓	ΔW4 ↓	Δabs-max ↓	Δp99.9 ↓	Δp99 ↓
Persistence	naïve	0.315	–	–	–	–	–	–
HAR	econometric	0.408	–	–	–	–	–	–
DLinear	linear	0.026±0.036	+0.0006	+0.0007	+0.0004	**+0.006±0.030**	+0.008±0.012	+0.024±0.018
PatchTST	attention	0.181±0.051	+0.0001	+0.0000	-0.0018	+0.019±0.041 (11%)	**+0.006±0.028 (3%)**	+0.008±0.022 (5%)
iTransformer	attention	0.197±0.060	+0.0004	+0.0002	+0.0030	+0.023±0.025 (12%)	+0.016±0.017 (8%)	**+0.015±0.017 (8%)**
TSMixer	mixer	0.490±0.043	+0.0002	+0.0002	+0.0020	+0.096±0.041 (20%)	**+0.020±0.012 (4%)**	+0.027±0.030 (5%)
Transformer	attention	0.490±0.048	+0.0001	+0.0003	+0.0004	+0.123±0.066 (25%)	+0.010±0.007 (2%)	**+0.007±0.008 (1%)**
SegRNN	recurrent	0.458±0.039	+0.0012	+0.0006	+0.0116	+0.271±0.068 (59%)	**+0.072±0.030 (16%)**	+0.107±0.046 (23%)
TimeMixer	mixer	0.303±0.049	+0.0036	+0.0084	+0.0075	+0.188±0.134 (62%)	+0.180±0.155 (59%)	**+0.095±0.096 (31%)**

iTransformer lose approximately 1.5%. As with W8A8, this figure is inflated for DLinear by its small denominator. No architecture loses more than 2.5%. TimeMixer and SegRNN show the largest weight-only effects, both around 2.5%. These results indicate that neither 8-bit activation quantization nor 4-bit weight quantization accounts for the substantial degradations observed later.

Result 2: Static 4-bit activation quantization causes substantial predictive losses. Under default abs-max calibration, PatchTST loses 11% of its FP32 signal and iTransformer loses 12%. The corresponding losses increase to 20% for TSMixer, 25% for Transformer, and approximately 60% for both TimeMixer and SegRNN. These differences do not follow a simple architecture taxonomy: attention-based models occupy several positions in the damage ordering, as do mixer-based models, so architecture family alone is insufficient to predict robustness under static W4A4.

The practical impact becomes clear when compared with the classical forecasting baselines. Transformer falls from an FP32 IC of 0.490 to 0.367 under default W4A4, placing it below HAR. SegRNN falls from 0.458 to 0.187, below the persistence benchmark. The best FP32 neural models exceed HAR by about 0.08 IC, while default W4A4 damage is 0.096 for TSMixer and 0.123 for Transformer. For these two architectures, quantization removes an amount of ranking accuracy comparable to, or larger than, the neural model's entire advantage over HAR. In contrast, DLinear, whose FP32 IC is close to zero, changes little under W4A4 and serves as a low-signal control.

However, for the architectures that are affected, percentile calibration (Table 2, p99.9 and p99 columns) substantially reduces damage in some cases but leaves large losses in others. This pattern motivates Section 6, which examines how activation-range selection shapes W4A4 damage. The magnitude of W4A4 damage also varies substantially across deployment periods, a pattern that Section 7 examines in detail.

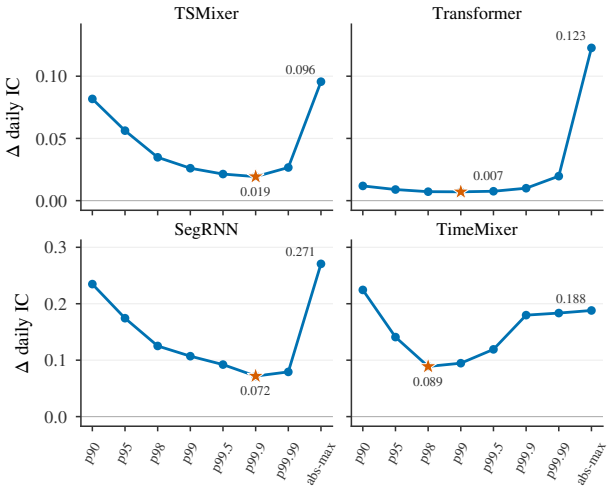


Figure 2: Activation-range sweep for W4A4, averaged over eight walk-forward folds and ten seeds per fold. Moving left narrows the range, improving in-range resolution while increasing clipping. The star marker indicates the lowest-damage setting for each architecture.

6 Why Default 4-Bit Calibration Fails

6.1 The range–resolution trade-off

A finite-range uniform quantizer creates two errors. Values inside the range are rounded. Values outside the range are clipped [2]. At 4 bits, the activation range decides which error dominates: a wider range reduces clipping but coarsens all in-range values, while a narrower range improves in-range resolution but clips more tail values.

Figure 2 shows this trade-off. We focus this analysis on the four architectures with the largest default W4A4 damage: TSMixer, Transformer, SegRNN, and TimeMixer. For all four, default abs-max calibration lies on the high-damage side of the curve. Damage first decreases as the range narrows, reaching a minimum somewhere between p98 and p99.9 depending on the architecture. For TSMixer, Transformer, and SegRNN, damage then rises sharply over the final step to abs-max. TimeMixer instead rises more gradually from around p99 onward, without a comparable late jump.

The curve also separates two cases. For TSMixer and Transformer, most of the default W4A4 failure is range-recoverable. For SegRNN and TimeMixer, changing the range helps but does not solve the problem. Those models remain sensitive even under the best tested range.

6.2 Recoverable and residual damage

Percentile calibration changes only the statistic used to set the activation range. The trained checkpoint, weights, calibration sample, and test observations stay fixed. Any improvement can therefore be attributed to range selection under the same quantizer. Transformer damage falls from +0.123 under abs-max to +0.007 at the best tested percentile, while TSMixer damage falls from +0.096 to +0.019.

We measure how much of the default abs-max damage can be recovered simply by switching the calibration statistic. Let $\Delta_{\text{abs-max}}$ denote an architecture's W4A4 damage under abs-max calibration, and let Δ_p denote its damage under percentile calibration at setting p (e.g., p99, p99.9). The best percentile setting is the one that minimizes Δ_p across the values tested in Figure 2, denoted $\min_p \Delta_p$. We define two quantities from this comparison: the *range-recoverable share* R, the fraction of abs-max damage eliminated by switching to the best percentile, and the *residual damage* L, the damage that remains even under that best setting. Formally,

$$R = \frac{\Delta_{\text{abs-max}} - \min_p \Delta_p}{\Delta_{\text{abs-max}}}, \qquad L = \min_p \Delta_p.$$

Because the best percentile here is selected using the same test-year outcomes it is evaluated on, R and L describe an upper bound on what range selection could achieve: a real deployment would have to choose the percentile in advance, using only data available before the test year begins. Section 8 turns these patterns into deployment guidance and tests whether that guidance survives without foresight of the test period.

The range-recoverable share is 94% for Transformer, 80% for TSMixer, 73% for SegRNN, and 53% for TimeMixer. These figures confirm the pattern in Figure 2: Transformer and TSMixer recover most of their default damage through range selection alone, while SegRNN and TimeMixer recover less than three-quarters and just over half, respectively.

6.3 Layerwise attribution

Range selection is not the only lever for reducing W4A4 damage. A layerwise check shows that Transformer and TSMixer respond very differently to precision reallocation at the layer level. For Transformer, one layer dominates: quantizing only the convolutional token embedding reproduces 72% of full-model W4A4 damage. Keeping that layer at 8 bits while quantizing the remaining eligible operations to 4 bits reduces damage from +0.123 to +0.015, essentially eliminating the W4A4 damage. TSMixer has no comparable single-layer bottleneck: protecting any individual layer leaves damage near +0.086, still large enough to erase most of the neural model's advantage over HAR.

7 Market Regime Sensitivity

Percentile calibration reduces average damage by exchanging some tail coverage for finer bulk resolution. This section examines how

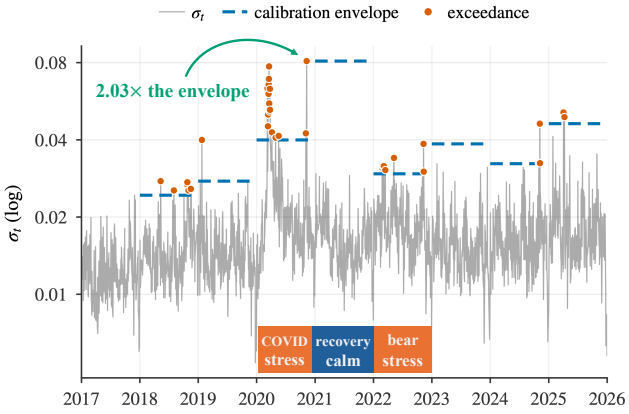


Figure 3: Daily cross-sectional return dispersion σ_t (gray) against the calibration envelope (dashed blue) for each walk-forward fold. Each envelope segment is the maximum σ_t observed during the validation year that precedes it and is held fixed while evaluating the following test year (orange markers indicate days that exceed it).

that trade-off changes when the market a model is tested on differs from the market it was calibrated on, using the COVID-19 shock of 2020, the calmer recovery in 2021, and the 2022 inflation and monetary-tightening episode as cases where this mismatch runs in opposite directions.

7.1 Calibration–test mismatch

Calibration always uses the most recent available data, so it can lag behind the market it is meant to prepare for. We show that the direction of that lag matters: a test year can be either more volatile or calmer than the year it was calibrated on, and these two directions damage the model in different ways.

We measure this lag using σ_t, the daily cross-sectional standard deviation of stock returns: since the models rank stocks by future volatility within each date, cross-sectional dispersion is a direct measure of how unusual that day's ranking environment is. For each fold, the *calibration envelope* is the largest σ_t observed during the preceding validation year. A test day lies outside the envelope when its dispersion exceeds that maximum.

Figure 3 shows that the envelope tracks recent history rather than a fixed baseline, and that 2020 and 2021 sit on opposite sides of it. In 2020, the COVID-19 shock pushes dispersion repeatedly beyond its own envelope: the market moves well beyond what the preceding year prepared the model for. In 2021, dispersion instead sits well below the envelope: the market recovers from the 2020 shock, and cross-sectional dispersion falls back toward typical levels, but the envelope still reflects the extremes of the year it was calibrated on. This asymmetry, being calibrated too narrow in one direction and too wide in the other, is what produces the two failure modes examined next.

This mismatch produces two opposite failure modes (Figure 4). When the test year is more volatile than its calibration year, as in 2020 and 2022, narrow percentiles cost more than usual: they clip

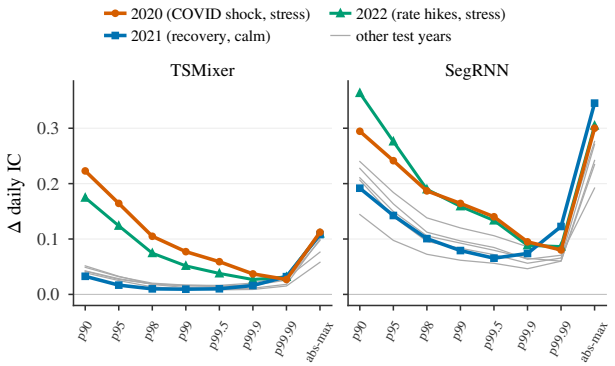


Figure 4: The activation-range sweep of Figure 2, separated by test year for TSMixer and SegRNN, against a baseline of all other test years (gray).

tail activations the calibration period underestimated. When the test year is calmer than its calibration year, as in 2021, the opposite happens: abs-max becomes the costliest setting, since it preserves resolution for extremes that no longer occur. For SegRNN, this reversal is severe enough that the 2021 fold has the largest abs-max damage of any fold in the panel.

This mismatch also inflates the damage that remains even after choosing the best percentile. Stress years produce more extreme tail activations, and no single percentile can fully cover them without sacrificing resolution elsewhere, so clipping-driven damage persists even under the best tested range. For TSMixer, residual damage in 2020 and 2022 is roughly twice that of 2021 and the remaining calmer folds.

7.2 Matched recalibration

We ask what would have happened to the 2020 fold if its calibration period had not been mismatched with its test period. The 2020 fold differs from other folds in several respects simultaneously: training sample, checkpoint, calibration period, and test period, so it is not possible to attribute its excess damage to calibration–test mismatch alone from the original comparison. To isolate this factor, we construct a matched comparison in which the 2020 fold is recalibrated on 2020 data itself, rather than on 2019, while the checkpoint, test days, range statistic, and sampling procedure remain unchanged.

This comparison is diagnostic rather than deployable, since it uses 2020 data that would not have been available before the year began. It answers a narrower question: how much of the 2020 damage under p99 calibration would have disappeared had calibration matched the test period, holding all other factors fixed.

Matched p99 recalibration reduces TSMixer damage from +0.077 to +0.043, a reduction of 44%, and SegRNN damage from +0.164 to +0.119, a reduction of 27%. Calibration–test mismatch therefore accounts for roughly a quarter to half of the 2020 damage under p99, depending on the architecture. The remaining loss persists even when calibration matches the test year exactly, indicating that it comes from clipping itself: a fixed percentile still cannot cover the extreme tail activations that a stress year like 2020 produces, no matter which year's data set that percentile uses.

Table 3: Averaged daily IC damage relative to FP32 inside and outside the calibration envelope. Positive values indicate lower IC after quantization. The ratio columns report outside/inside damage under p99 and abs-max calibration.

	Δp99			Δabs-max		
Model	Inside	Outside	Ratio	Inside	Outside	Ratio
TimeMixer	+0.094	+0.140	1.5	+0.187	+0.266	1.4
TSMixer	+0.026	+0.056	2.2	+0.096	+0.088	0.9
SegRNN	+0.104	+0.260	2.5	+0.270	+0.285	1.1
Transformer	+0.007	+0.025	3.6	+0.123	+0.095	0.8

Table 4: Deployment choices implied by the range and residual diagnostics.

Observed pattern	Suggested setting
No meaningful W4A4 damage	Keep the default range
Most damage disappears after range tuning	Percentile W4A4
One layer accounts for most damage	Keep that layer at 8 bits
Large damage remains across tested ranges	Dyn8, W8A8, or W4
Test conditions exceed calibration history	Recalibrate or use 8 bits

7.3 The cost of narrow ranges on extreme days

Percentile calibration lowers average damage, but Table 3 shows that this saving is conditional, not free. Under p99, damage outside the calibration envelope is 1.5–3.6× larger than inside, for all four architectures. The narrow range that helps on typical days clips more heavily once conditions turn extreme. Its advantage disappears when reliability matters most, on the unusual market days a deployed model is least prepared for. Abs-max does not carry this cost. It is already wide enough to cover extremes, so its damage outside the envelope is no higher than inside for TSMixer and Transformer, and modestly higher for TimeMixer and SegRNN.

8 Deployment Implications and Limitations

A deployment procedure. Table 4 summarizes the resulting deployment procedure. Start from the full-precision checkpoint and measure default W4A4 damage before deployment. If the damage is small, no range intervention is needed. If the damage is large, sweep percentile ranges on data available before deployment and compute R and L; together with the layerwise check, these determine the setting shown in the table. After deployment, the calibration envelope provides a monitoring signal: among the architectures most sensitive to activation range, out-of-envelope days have 1.5–3.6× the typical daily quantization damage under p99 calibration. Such days can trigger recalibration, a more conservative range such as p99.9, or a fallback to 8-bit activations.

Does this guidance survive without hindsight? The range sweeps in Section 6 choose the best percentile using test-year outcomes, so they are diagnostic rather than deployable. To test a deployable rule, we repeat the same sweep on the validation year.

For each architecture and fold, we choose the percentile p with the smallest validation-year damage Δ_p, then apply that percentile to the following test year. We refer to the retrospectively best test-year percentile as the *oracle* percentile.

This validation rule preserves the coarse guidance from R and L for all seven architectures. Across the four swept architectures, the validation-selected percentile has mean regret of 0.007 IC and median regret of 0.002 IC relative to the oracle, and matches or improves on a fixed p99 rule in 27 of the 32 architecture–fold cells. The largest failure occurs for SegRNN in the 2021 fold. Its percentile is selected from the 2020 validation year, a COVID-stress period, and then applied to the calmer 2021 test year. The resulting regret is 0.057 IC.

Limitations and future work. Reported IC levels are survivorship-tilted because the panel uses current S&P 500 constituents. This bias is shared across both conditions, since quantized models are compared with their own full-precision checkpoints on identical stocks and dates. Quantization is simulated in FP32, capturing the predictive cost of range selection but not backend-level efficiency or accumulation effects. We also study only symmetric round-to-nearest quantization, in its standard hardware-friendly layout: per-tensor activation ranges with per-channel weight scales. Stronger PTQ methods [1, 22] may shrink the residual damage of SegRNN and TimeMixer, an open question this framework can answer. Finally, the eight test years contain only two major stress episodes, and skill is measured by IC alone. Hardware validation, other markets, and portfolio-level evaluation remain future work.

9 Conclusion

Post-training quantization is not uniformly risky for financial time-series forecasting. 8-bit quantization and weight-only 4-bit quantization generally cause little damage. The large failures come from static 4-bit activation quantization, where default abs-max calibration removes a substantial share of predictive signal.

Activation-range selection is a major source of this damage. Percentile calibration recovers most of the loss, though recurrent and multi-scale architectures retain substantial residual damage, and the preferred range shifts across market regimes. Where a single layer dominates the damage, protecting it at 8 bits offers an alternative lever to range selection.

Calibration should therefore be treated as part of the deployment policy, not a fixed preprocessing step: measure default W4A4 damage, sweep the activation range, and fall back to 8-bit activations, weight-only W4, or a layer-wise exception when residual damage remains large. This guidance largely holds without foresight, though regime mismatch can still cause sizable regret.

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